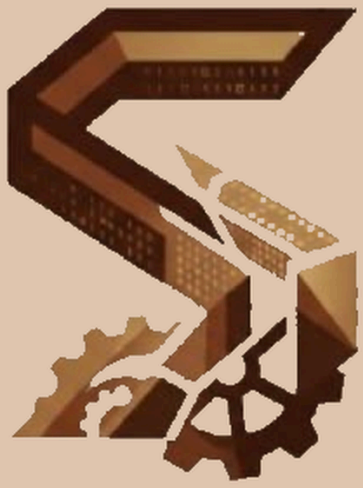


Spectrum
SPC

From Binary to Beyond

بنك أسئلة كالكولاس 1

الوحدتان 0 و 1 · 100 سؤال اختيار من متعدد

الوحدة 0: Before Calculus

أول 15 سؤالاً سهلة، ثم 20 متوسطة، وآخر 15 صعبة.

sqrt = الجذر التربيعي، arccosg تعني $\cos^{-1}$.

1 If f(x) = 2x + 3, then f(4) =

A 8
B 9
C 11
D 14

2 The domain of f(x) = sqrt(x - 2) is

A x > 2
B x ≥ 2
C x ≤ 2
D all real numbers

3 The domain of f(x) = 1/(x - 5) is

A x ≠ 0
B x > 5
C x ≠ -5
D x ≠ 5

4 If $f(x) = x^2 - 1$, then f(-3) =

A 8
B -10
C -8
D 10

5 The range of $f(x) = x^2$ is

A (-∞, ∞)
B (0, ∞)
C [0, ∞)
D (-∞, 0]

6 The function $y = x^3$ is

A even
B odd
C both even and odd
D neither even nor odd

7 The slope of the line through (1, 2) and (3, 8) is

A 2
B 4
C 6
D 3

8 The line with slope -2 passing through (0, 5) is

A $y = -2x + 5$
B $y = 2x + 5$
C $y = -2x - 5$
D $y = 5x - 2$

9 If $f(x) = x + 1$ and $g(x) = 2x$, then $(f \circ g)(x) =$

A $2x + 2$
B $x + 2$
C $2x + 1$
D $2x$

10 The solution of $|x - 3| < 2$ is

A $(-1, 5)$
B $(1, 5)$
C $(-5, 1)$
D $x < 1$ or $x > 5$

11 $\sin(\pi/6) =$

A $\sqrt{3}/2$
B 1
C $\sqrt{2}/2$
D $1/2$

12 $120°$ in radians equals

A $2\pi/3$
B $\pi/3$
C $\pi/2$
D $3\pi/4$

13 The vertex of $y = x^2 - 4x + 3$ is

A $(-2, -1)$
B $(4, 3)$
C $(2, -1)$
D $(2, 1)$

14 The domain of $f(x) = \sqrt{9 - x^2}$ is

A $(-3, 3)$
B $x \geq 3$
C $x \leq -3$ or $x \geq 3$
D $[-3, 3]$

15 Does $x^2 + y^2 = 1$ define y as a function of x?

A Yes
B No, it fails the vertical line test
C Only when $x \geq 0$
D Yes, because it is continuous

16 The inverse of $f(x) = 3x - 6$ is

A $(x + 6)/3$
B $(x - 6)/3$
C $3x + 6$
D $1/(3x - 6)$

17 $\log_2(8) =$

A 2
B 4
C 3
D 8

18 $\ln(e^5) =$

A e^5
B 5
C 1/5
D 5e

19 The solution of $2^x = 32$ is

A 4
B 6
C 16
D 5

20 Which function is even?

A $x^4 - x^2$
B $x^3 + x$
C x^3
D $x^4 + x$

21 For $f(x) = x^2$, $[f(x + h) - f(x)]/h =$

A $2x$
B $x + h$
C $2x + h$
D $2x + 2h$

22 The domain of $f(x) = \ln(x - 1)$ is

A $x \geq 1$
B $x > 0$
C $x < 1$
D $x > 1$

23 The domain of $f(x) = (x + 1)/(x^2 - 4)$ is

A $x \neq 2$
B $x \neq \pm 2$
C $x \neq \pm 4$
D $x \neq -1$

24 The graph of $y = -(x - 2)^2 + 1$ has

A vertex (2, 1) and opens downward
B vertex (-2, 1) and opens downward
C vertex (2, 1) and opens upward
D vertex (2, -1) and opens downward

25 The solution set of $x^2 - 5x + 6 \geq 0$ is

A [2, 3]
B (2, 3)
C $(-\infty, 2] \cup [3, \infty)$
D $x \geq 3$

26 If f(x) = sqrt(x) and $g(x) = x^2 - 4$, the domain of (f o g)(x) is

A [-2, 2]
B $|x| \geq 2$
C $x \geq 2$
D (-2, 2)

27 sin(5pi/6) =

A -1/2
B sqrt(3)/2
C -sqrt(3)/2
D 1/2

28 The solutions of sin x = 1/2 in [0, 2pi) are

A pi/6 and 5pi/6
B pi/6 and 7pi/6
C pi/3 and 2pi/3
D pi/6 only

29 The solution of log_3(x + 2) = 2 is

A 4
B 9
C 7
D 11

30 ln 8 - ln 2 =

A ln 6
B 4
C ln 16
D ln 4

31 The inverse of $f(x) = x^3 + 1$ is

A cube root of x, minus 1
B cube root of (x - 1)
C $(x - 1)^3$
D $1/(x^3 + 1)$

32 The inverse of $f(x) = (2x + 1)/(x - 3)$ is

A $(3x + 1)/(x - 2)$
B $(3x - 1)/(x + 2)$
C $(x - 3)/(2x + 1)$
D $(x + 3)/(2x - 1)$

33 Is $f(x) = x^2$ one-to-one?

A Yes, on all real numbers
B No; its inverse is x^2
C Yes; its inverse is $\pm\sqrt{x}$
D No on all reals, but on $[0, \infty)$ its inverse is $\sqrt{x}$

34 The solutions of $e^{2x} - 5e^x + 6 = 0$ are

A $x = 2, 3$
B $x = \ln 5$
C $x = \ln 2, \ln 3$
D $x = e^2, e^3$

35 $\arccos(-1/2) =$

A $\pi/3$
B $2\pi/3$
C $4\pi/3$
D $-\pi/3$

36 $\sin(\arccos(3/5)) =$

A $4/5$
B $3/5$
C $5/4$
D $3/4$

37 For $-1 < x < 1$, $\tan(\arcsin x) =$

A $x/\sqrt{1 + x^2}$
B $\sqrt{1 - x^2}/x$
C $x\sqrt{1 - x^2}$
D $x/\sqrt{1 - x^2}$

38 The solution of $\ln x + \ln(x - 2) = \ln 3$ is

A -1
B 5
C 3
D -1 and 3

39 Let $f(x) = x + 1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Then $f(-2) + f(3) =$

A 6
B 8
C 10
D 12

40 The domain and range of $f(x) = 3 - \sqrt{x + 1}$ are

A domain $[-1, \infty)$, range $(-\infty, 3]$
B domain $(-1, \infty)$, range $[3, \infty)$
C domain $[-1, \infty)$, range $[3, \infty)$
D domain $[0, \infty)$, range $(-\infty, 3]$

41 The range of $f(x) = x^2 - 4x + 7$ is

A $[7, \infty)$
B $(-\infty, 3]$
C $[-3, \infty)$
D $[3, \infty)$

42 The solution of $|2x - 1| \geq 5$ is

A $-2 \leq x \leq 3$
B $x \leq -3$ or $x \geq 2$
C $x \leq -2$ or $x \geq 3$
D $x \geq 3$

43 The solutions of $2\cos^2 x - \cos x - 1 = 0$ in $[0, 2\pi)$ are

A 0 and $2\pi/3$
B $0, 2\pi/3, 4\pi/3$
C $2\pi/3$ and $4\pi/3$
D $\pi/3$ and $5\pi/3$

44 The inverse of $f(x) = x/(x + 1)$ is

A $x/(1 - x)$
B $x/(1 + x)$
C $(1 - x)/x$
D $-x/(x + 1)$

45 The inverse of $f(x) = 1 + \ln(x - 2)$ is

A $1 + e^{x - 2}$
B $2 - e^{x - 1}$
C $2 + e^{x - 1}$
D $e^{x + 1} - 2$

46 The even part of e^x is

A e^{-x}
B $(e^x - e^{-x})/2$
C $e^x/2$
D $(e^x + e^{-x})/2$

47 $\sin(2 \arctan(3/4)) =$

A $12/25$
B $24/25$
C $7/25$
D $3/5$

48 A population $P = 200\,e^{0.03t}$ reaches 1000 when $t \approx$

A 53.6
B 5.4
C 167
D 33.3

49 The domain of $f(x) = \ln(x^2 - 4x + 3)$ is

A $(1, 3)$
B $[1, 3]$
C $x < 1$ or $x > 3$
D $x \neq 1, 3$

50 The function $f(x) = \ln(x + \sqrt{x^2 + 1})$ is

A even
B neither even nor odd
C periodic
D odd

الوحدة 1: Limits and Continuity

أول 15 سؤالاً سهلة، ثم 20 متوسطة، وآخر 15 صعبة.

51 $\lim_{x \to 2} (3x + 1) =$

A 5
B 7
C 6
D 9

52 $\lim_{x \to 3} x^2 =$

A 6
B 3
C 27
D 9

53 $\lim_{x \to -1} (x^2 - 2x + 4) =$

A 7
B 3
C 1
D 5

54 $\lim_{x \to 5} 7 =$

A 5
B 0
C 7
D does not exist

55 $\lim_{x \to 2} (x^2 - 4)/(x - 2) =$

A 0
B 2
C does not exist
D 4

56 $\lim_{x \to 0} (x^2 + 3x)/x =$

A 3
B 0
C 1
D does not exist

57 $\lim_{x \to 1} (x + 2)/(x + 3) =$

A 1
B 2/3
C 3/4
D 1/4

58 $\lim_{x \to 0} \cos x =$

A 0
B 1
C -1
D does not exist

59 $\lim_{x \to +\infty} 1/x =$

A 0
B 1
C $+\infty$
D does not exist

60 $\lim_{x \to 4} \sqrt{x} =$

A 4
B 16
C 2
D -2

61 $\lim_{x \to 3} (x^2 - 9)/(x - 3) =$

A 3
B 6
C 9
D 0

62 $\lim_{x \to -2} (x^2 + x - 2)/(x + 2) =$

A 0
B 3
C -1
D -3

63 $\lim_{x \to 0} (\sin x)/x =$

A 0
B $+\infty$
C 1
D does not exist

64 $\lim_{x \to 1} (x^2 - 1)/(x^2 - 3x + 2) =$

A 2
B -2
C 0
D 1/2

65 $\lim_{x \to 0} 1/x$

A $= +\infty$
B $= -\infty$
C $= 0$
D does not exist

66 $\lim_{x \to 0} [\sqrt{x + 4} - 2]/x =$

A 1/4
B 1/2
C 0
D 4

67 $\lim_{x\to+\infty} (3x^2 + 2x)/(5x^2 - 1) =$

A 0
B 3/5
C 5/3
D +∞

68 $\lim_{x\to+\infty} (2x + 1)/(x^2 + 3) =$

A 0
B 2
C +∞
D 1/3

69 $\lim_{x\to0} \sin(3x)/x =$

A 1
B 1/3
C 0
D 3

70 $\lim_{x\to0} (\tan x)/x =$

A 0
B +∞
C 1
D does not exist

71 $\lim_{x\to0} (1 - \cos x)/x =$

A 1
B 1/2
C +∞
D 0

72 $\lim_{x\to3+} 1/(x - 3) =$

A -∞
B 0
C +∞
D 1

73 $\lim_{x\to-\infty} (x^3 + 1)/(x^2 + 1) =$

A -∞
B +∞
C 0
D 1

74 $\lim \operatorname{sqrt}(x^2 + 1)/x$ equals

A 1 as $x\to\pm\infty$
B 1 as $x\to+\infty$ and -1 as $x\to-\infty$
C -1 as $x\to\pm\infty$
D 0 as $x\to\pm\infty$

75 $\lim_{x\to 0} \sin(5x)/\sin(2x) =$

A 5/2
B 2/5
C 1
D 10

76 $\lim_{x\to 0} \sin(4x)/\tan(3x) =$

A 3/4
B 4/3
C 12
D 1

77 $\lim_{x\to 0} (1 - \cos 2x)/x^2 =$

A 1/2
B 4
C 2
D 1

78 $\lim_{x\to 1} (x^3 - 1)/(x - 1) =$

A 1
B 0
C 2
D 3

79 $f(x) = (x + 1)/(x^2 - x - 6)$ is discontinuous at

A $x = 3$ only
B $x = -3$ and $x = 2$
C $x = 3$ and $x = -2$
D $x = -1$

80 $f(x) = (x^2 - 4)/(x - 2)$ for $x \neq 2$, with $f(2) = k$, is continuous at $x = 2$ when $k =$

A 0
B 2
C -4
D 4

81 $\lim_{x\to 0} |x|/x$

A = 1
B does not exist
C = -1
D = 0

82 For $f(x) = 2x + 1$ $(x < 3)$ and $f(x) = ax - 2$ $(x \geq 3)$ to be continuous at $x = 3$, $a =$

A 3
B 7/3
C 5
D 1

83 The vertical asymptotes of $f(x) = (x + 1)/(x^2 - 4)$ are

A $x = 2$ only
B $x = 2$ and $x = -2$
C $x = -1$
D there are none

84 The horizontal asymptote of $f(x) = (2x^2 + 1)/(x^2 - 3)$ is

A $y = 0$
B $y = 1$
C $y = 2$
D $y = -1$

85 $\lim (x \to +\infty) [\sqrt{(x^2 + 3x)} - x] =$

A 0
B 3
C $+\infty$
D 3/2

86 $\lim (x \to 0) [\sqrt{(1 + x)} - \sqrt{(1 - x)}]/x =$

A 1
B 0
C 2
D 1/2

87 $\lim (x \to -\infty) (3x^3 - x)/(2x^3 + 5x^2) =$

A 0
B 3/2
C -3/2
D $+\infty$

88 By the Squeeze Theorem, $\lim (x \to 0)\ x \sin(1/x) =$

A 1
B does not exist
C $+\infty$
D 0

89 By the Squeeze Theorem, $\lim (x \to +\infty) (\sin x)/x =$

A 1
B does not exist
C 0
D $+\infty$

90 $\lim (x \to 0+) \ln x =$

A $+\infty$
B $-\infty$
C 0
D 1

91 $\lim_{x\to 0} \sin x/(x^2 + x) =$

A 0
B $+\infty$
C 1/2
D 1

92 By the Intermediate Value Theorem, $x^3 + x - 1 = 0$ must have a root in

A (0, 1)
B (1, 2)
C (-1, 0)
D (2, 3)

93 By the Intermediate Value Theorem, $\cos x = x$ must have a solution in

A (-1, 0)
B (1, 2)
C (0, 1)
D (2, 3)

94 To prove $\lim_{x\to 2} (3x - 1) = 5$ with the ε-δ definition, a suitable choice is $\delta =$

A ε
B $\varepsilon/3$
C 3ε
D $\varepsilon/5$

95 For $\lim_{x\to 2} x^2 = 4$ with $\varepsilon = 0.1$ and $\delta \le 1$, we have $|x^2 - 4| < 5\delta$. The largest δ that guarantees $|x^2 - 4| < 0.1$ is

A 0.02
B 0.1
C 0.5
D 0.2

96 To prove $\lim_{x\to 2} x^2 = 4$ for any $\varepsilon > 0$, a valid choice is $\delta =$

A $\varepsilon/2$
B $\min(1, \varepsilon)$
C $\min(1, \varepsilon/5)$
D $\varepsilon/4$

97 $f(x) = (x^2 - 1)/(x - 1)$ for $x \neq 1$ and $f(1) = 3$ has at $x = 1$

A a jump discontinuity
B an infinite discontinuity
C an essential discontinuity
D a removable discontinuity

98 f is continuous everywhere, where $f(x) = 2$ for $x \le -1$; $ax + b$ for $-1 < x < 3$; -4 for $x \ge 3$. Then

A $a = -3/2, b = 1/2$
B $a = 3/2, b = -1/2$
C $a = -2, b = 0$
D $a = -1/2, b = 3/2$

99 $\lim_{x\to 0} (1 - \cos x)/x^2 =$

A 1
B 1/2
C 0
D 2

100 $\lim_{x\to 0} (\tan x - \sin x)/x^3 =$

A 1
B 0
C 1/2
D 2

مفتاح الإجابات

1	C	2	B	3	D	4	A	5	C	6	B	7	D	8	A
9	C	10	B	11	D	12	A	13	C	14	D	15	B	16	A
17	C	18	B	19	D	20	A	21	C	22	D	23	B	24	A
25	C	26	B	27	D	28	A	29	C	30	D	31	B	32	A
33	D	34	C	35	B	36	A	37	D	38	C	39	B	40	A
41	D	42	C	43	B	44	A	45	C	46	D	47	B	48	A
49	C	50	D	51	B	52	D	53	A	54	C	55	D	56	A
57	C	58	B	59	A	60	C	61	B	62	D	63	C	64	B
65	D	66	A	67	B	68	A	69	D	70	C	71	D	72	C
73	A	74	B	75	A	76	B	77	C	78	D	79	C	80	D
81	B	82	A	83	B	84	C	85	D	86	A	87	B	88	D
89	C	90	B	91	D	92	A	93	C	94	B	95	A	96	C
97	D	98	A	99	B	100	C								